European Price under LLH Dynamics

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DISCLAIMER

ACKNOWLEDGMENT

1 Introduction

We investigate the semi analytical formula described in the paper of Lin, Lin, and He [1] for the price of a European option written on an underlying whose price process is assumed to follow an improved Stein-Stein two factor stochastic volatility model. This model is an improvement over the Stein-Stein model introduced in [2]. The authors in [1] allow the brownian motion risk drivers to be correlated and the long-run mean value of volatility to depend on time and to be stochastic.

In an attempt to implement the semi-analytical formula in [1] that is based on Theorem 3.2 in this document, we found it compulsory to re-derive the system of ODEs in [1] which may be found in section 4 of this document. The theorem and the ODE system allow the computation of the European price using the characteristics f_j of the log price process and its corresponding distribution functions P_j under risk neutral measures as described in section 3.1.

While investigating the derivation of the European price formula in [1], , we stumbled upon the work of Schobel and Zhu [3] in which the authors consider the Stein-Stein model [2] while allowing correlated risk drivers (the difference with [1] is that the long-run mean of the volatility remains constant). The approaches in [1] and [3] are different. One that is **PDE based** following **Heston** [4], and the other is **stochastic calculus** based. In [1], the authors follow the former approach which using a standard martingale argument, a PDE for U is derived that ultimately yields PDEs for the $P_j's$ and consequently the $f_j's$. Following the ansatz from Lord and Kahl [5] that for affine diffusion models the $f_j's$ must be exponential in the state variables, leads to the system of ODEs described in [1] and that may be found in section 4 of this document.

In [3] the authors apply stochastic calculus machinery to derive the $f_j(\phi)$'s in terms of the model variables and parameters. We found their approach more modern and more transparent. While adapting their approach to the LLH model, we found that there is a contradiction with the assumption of an "affine diffusion model" and we found that the characteristic function $f_1(\phi)$ cannot be assumed the exponential of a quadratic function of the volatility and it long-run mean as was obtained in [1].

2 The Model of the Price Dynamics

The price dynamics under the Lin-Lin-He (LLH) model [1] is given by the following system of SDEs:

$$\frac{dS_t}{S_t} = rdt + \sigma_t dW_t^1 \tag{1}$$

$$d\sigma_t = \kappa(\theta_t - \sigma_t)dt + \nu dW_t^2 \tag{2}$$

$$d\theta_t = \lambda dt + \eta dB_t \tag{3}$$

Where W_t^1 and W_t^2 are standard Brownian motions with correlation ρ, B_t is a standard geometric Brownian motion, and θ_t is the long-run mean value of volatility on a specific time interval t.

Remark 2.1.

(i) $[W_t^1, W_t^2] = \text{cov}\,(W_t^1, W_t^2) = \rho t$, where the brackets $[\cdot, \cdot]$ operator is that of quadratic covariation.

(ii) B_t is given by the following SDE:

$$dB_t = B_t dW_t, \quad B_0 = 1 \iff B_t = \exp\left(W_t - \frac{1}{2}t\right) \tag{4}$$

where W_t is a standard Brownian motion that is independent of W_t^1 and W_t^2.

Proposition 2.2. *For all $t \geq 0$,*

$$\theta_t = \theta_0 + \lambda t + \eta(B_t - B_0) \tag{5}$$

$$\begin{aligned}\sigma_t = {} & e^{-\kappa t}(\sigma_0 + \lambda/\kappa - \theta_0) + \theta_0 + \lambda(t - 1/\kappa) + \nu W_t^2 - \\ & \nu\kappa \int_0^t e^{-\kappa s} W_{t-s}^2 ds + \eta\kappa \int_0^t e^{-\kappa s}(B_{t-s} - B_0) ds\end{aligned} \tag{6}$$

$$\begin{aligned}\textit{alternative}\ \sigma_t = {} & e^{-\kappa t}(\sigma_0 + \lambda/\kappa - \theta_0) + \theta_0 + \lambda(t - 1/\kappa) + \\ & \eta(B_t - B_0) + e^{-\kappa t}\left(\eta \int_0^t e^{\kappa u} dB_u + \nu \int_0^t e^{\kappa u} dW_u^2\right)\end{aligned} \tag{7}$$

Proof. A direct integration of (3) yields

$$\theta_t = \theta_0 + \lambda t + \eta(B_t - B_0)$$

To integrate Eq. (2), proceed with the preliminary computation:

$$\begin{aligned}d(e^{\kappa t}\sigma_t) &= \kappa e^{\kappa t}\sigma_t dt + e^{\kappa t} d\sigma_t \\ &= \kappa e^{\kappa t}\sigma_t dt + \kappa e^{\kappa t}(\theta_t - \sigma_t) dt + \nu e^{\kappa t} dW_t^2 \\ &= \kappa e^{\kappa t}\theta_t dt + \nu e^{\kappa t} dW_t^2\end{aligned}$$

Thus, direct integration yields σ_t in the following equation:

$$\begin{aligned} e^{\kappa t}\sigma_t - \sigma_0 &= \kappa \int_0^t e^{\kappa u}\left[\theta_0 + \lambda u + \eta(B_u - B_0)\right] du + \nu \int_0^t e^{\kappa u} dW_u^2 \\ &= \theta_0(e^{\kappa t} - 1) + \lambda\kappa \int_0^t u e^{\kappa u} du + \eta\kappa \int_0^t e^{\kappa u}(B_u - B_0) du + \nu \int_0^t e^{\kappa u} dW_u^2 \end{aligned}$$

Integration by parts simplifies the equation to

$$\begin{aligned} e^{\kappa t}\sigma_t - \sigma_0 &= \theta_0(e^{\kappa t} - 1) + \lambda\left[e^{\kappa t}(t - 1/\kappa) + 1/\kappa\right] + \eta\kappa \int_0^t e^{\kappa u}(B_u - B_0) du \\ &\quad + \nu e^{\kappa t} W_t^2 - \nu\kappa \int_0^t e^{\kappa u} W_u^2 du \\ &= \lambda/\kappa - \theta_0 + \theta_0 e^{\kappa t} + \lambda(t - 1/\kappa)e^{\kappa t} + \nu e^{\kappa t} W_t^2 - \\ &\quad \nu\kappa \int_0^t e^{\kappa u} W_u^2 du + \eta\kappa \int_0^t e^{\kappa u}(B_u - B_0) du. \end{aligned}$$

Thus,

$$\begin{aligned} \sigma_t &= e^{-\kappa t}(\sigma_0 + \lambda/\kappa - \theta_0) + \theta_0 + \lambda(t - 1/\kappa) + \nu W_t^2 - \\ &\quad \nu\kappa \int_0^t e^{-\kappa(t-u)} W_u^2 du + \eta\kappa \int_0^t e^{-\kappa(t-u)}(B_u - B_0) du \end{aligned}$$

which yields (6) after a change of variable of $s = t - u$ in the integral terms. □

Proposition 2.3. *For $t \geq 0$:*

$$S_t = S_0 \exp\left(rt - \frac{1}{2}\int_0^t \sigma_t^2 dt + \int_0^t \sigma_t dW_t^1\right) \tag{8}$$

amenable to bias-variance analysis

Proof Apply Ito's lemma to the log price process $X_t = \log S_t$ to obtain :

$$\begin{aligned} dX_t &= \frac{1}{S_t} dS_t - \frac{1}{2S_t^2} dS_t \cdot dS_t \\ &= \frac{dS_t}{S_t} - \frac{1}{2S_t^2} \sigma_t^2 S_t^2 dt \\ &= \left(r - \frac{1}{2}\sigma_t^2 \right) dt + \sigma_t dW_t^1 \end{aligned}$$

Thus,

$$X_t = X_0 + rt - \frac{1}{2}\int_0^t \sigma_u^2 du + \int_0^t \sigma_u dW_u^1 \tag{9}$$

which yields (8) when X_t is inserted into $S_t = e^{X_t}$ □

Remark 2.4.

(i) Note that for $\eta = 0$:

$$\begin{aligned} \sigma_t &= e^{-\kappa t}(\sigma_0 + \lambda/\kappa - \theta_0) + \theta_0 + \lambda(t - 1/\kappa) + \nu e^{-\kappa t} \int_0^t e^{\kappa u} dW_u^2 \\ &\sim \mathcal{N}\left(e^{-\kappa t}(\sigma_0 + \lambda/\kappa - \theta_0) + \theta_0 + \lambda(t - 1/\kappa),\ \frac{\nu^2}{2\kappa}(1 - e^{-2\kappa t}) \right) \end{aligned}$$

(ii) Notice that if the interval $[0, t]$ is discretized to $n + 1$ points using a time step Δt, formula (6) yields the following estimator for σ_t:

$$\begin{aligned} \hat{\sigma}_t = e^{-\kappa t}(\sigma_0 + \lambda/\kappa - \theta_0) + \theta_0 + \lambda(t - 1/\kappa) + \nu W_t^2 - \\ \nu\kappa\Delta t \sum_{i=1}^{n} e^{-\kappa i \Delta t} W_{t-i\Delta t}^2 + \eta\kappa\Delta t \sum_{i=1}^{n} e^{-\kappa i \Delta t}(B_{t-i\Delta t} - B_0) \end{aligned} \tag{10}$$

$$\text{alternative } \hat{\sigma}_t \approx e^{-\kappa t}(\sigma_0 + \lambda/\kappa - \theta_0) + \theta_0 + \lambda(t - 1/\kappa) + \eta(B_t - B_0)$$
$$+ e^{-\kappa t}\left(\eta \sum_{i=1}^{n} e^{\kappa(i-1)\Delta t}\Delta B_i + \nu \sum_{i=1}^{j} e^{\kappa(i-1)\Delta t}\Delta W_i^2\right) \tag{11}$$

where ΔB_i and ΔW_i^2 are increments of the processes B_t and W_t^2 on the interval $[(i-1)\Delta t, i\Delta t]$, $i = 1, \ldots, n$ Notice the variance of $\hat{\sigma}_t$ may be computed, so may be useful in investigating convergence in MSE. Also, it may be worthwhile to review the estimates of the stochastic integrals.

(iii) Similarly, formulas (9) and (8) yield the following estimators of the price:

$$\hat{S}_t = S_0 e^{rt} \exp\left(-\frac{\Delta t}{2}\sum_{i=0}^{n-1}\sigma_i^2 + \sum_{i=0}^{n-1}\sigma_i \Delta W_i^1\right) \tag{12}$$

2.1 Simulating Paths from the Price Process

Given the parameters κ, ν, λ, η, and ρ, the price process path may be simulated by following the algorithm:

Algorithm 2.5.

To simulate a price path up to time t, disrectize time to $n+1$ nodes $t_j = j\Delta t$ with time step Δt and $j = 0, \ldots, n$.

(i) Generate n i.i.d. samples Z_i from a standard normal. Set $\Delta W_i = \sqrt{\Delta t} Z_i$ and notice that

$$W_j := W_{t_j} = \sum_{i=1}^{j} \Delta W_i = \sqrt{\Delta t}\sum_{i=1}^{j} Z_i$$

then use Eq. (4) to compute B_t and its increments as follows:

$$B_j := B_{t_j} = e^{-\frac{j\Delta t}{2}} \exp W_j \tag{13}$$

$$\Delta B_j := B_j - B_{j-1} \tag{14}$$

for $j = 1, \ldots, n$.

(ii) Generate independently n i.i.d. samples Z_i^1 and Z_i^2 from a standard normal. Set the increments:

$$\Delta W_i^1 := \sqrt{\Delta t} Z_i^1 \text{ and } \Delta W_i^2 := \sqrt{\Delta t}\left(\rho Z_i^1 + \sqrt{1-\rho^2} Z_i^2\right) \text{ for } i = 1, \ldots, n$$

then compute

$$W_j^2 := W_{t_j}^2 = \sum_{i=1}^{j} \Delta W_i^2 = \sqrt{\Delta t} \sum_{i=1}^{j} \left(\rho Z_i^1 + \sqrt{1-\rho^2} Z_i^2\right) \text{ for } j = 1, \ldots, n$$

(iii) For $j = 1, \ldots, n$ use Eqs. (10), (13), and (14) to estimate the volatilities:

$$\begin{aligned}
\hat{\sigma}_j &:= \hat{\sigma}_{t_j} \\
&\approx e^{-\kappa j \Delta t}(\sigma_0 + \lambda/\kappa - \theta_0) + \theta_0 + \lambda(j\Delta t - 1/\kappa) + \nu W_j^2 - \\
&\quad \nu\kappa\Delta t \sum_{i=1}^{j} e^{-\kappa i \Delta t} W_{j-i}^2 + \eta\kappa\Delta t \sum_{i=1}^{j} e^{-\kappa i \Delta t}(B_{j-i} - 1),
\end{aligned} \tag{15}$$

Notice that these estimates have been obtained as Riemann sum approximations of the definite integrals. One can well use more accurate and more efficient numerical integration methods (quadrature, guass-). Mind the exponents of Δt in the estimation

of the volatility! Analyze the exponentially decaying terms as $\Delta t \to 0$.

$$\begin{aligned}
\text{alternative}\hat{\sigma}_j &:= \hat{\sigma}_{t_j} \\
&\approx e^{-\kappa j \Delta t}(\sigma_0 + \lambda/\kappa - \theta_0) + \theta_0 + \lambda(j\Delta t - 1/\kappa) + \eta(B_j - 1) \\
&\quad + e^{-\kappa j \Delta t}\left(\eta \sum_{i=1}^{j} e^{\kappa(i-1)\Delta t}\Delta B_i + \nu \sum_{i=1}^{j} e^{\kappa(i-1)\Delta t}\Delta W_i^2\right)
\end{aligned}$$

(iv) For $j = 1, \ldots, n$ use Eq. (8) to estimate the price values:

$$\begin{aligned}
\hat{S}_j &= \hat{S}_{t_j} \\
&= S_0 e^{rj\Delta t} \exp\left(-\frac{\Delta t}{2}\sum_{i=0}^{j-1}\sigma_i^2 + \sum_{i=0}^{j-1}\sigma_i \Delta W_i^1\right)
\end{aligned} \tag{16}$$

alternative Solve the following recurrence equation for $\hat{S}_j$, $j = 1, \ldots, n$:

$$\hat{S}_{i+1} - \hat{S}_i = \hat{S}_i\left(r\Delta t + \hat{\sigma}_i \Delta W_i^1\right) \tag{17}$$

Notice that the recurrence relation yields:

$$\begin{aligned}
\hat{S}_j &= S_0 \prod_{i=0}^{j-1}\left(1 + r\Delta t + \hat{\sigma}_i \Delta W_i^1\right) \\
&= S_0 \prod_{i=0}^{j-1}\left(1 + r\Delta t + \hat{\sigma}_i \sqrt{\Delta t} Z_i^1\right)
\end{aligned} \tag{18}$$

Can I consider a more accurate discetization scheme of the price?

Remark 2.6. Notice that (13), (15), and (18) suggest the terms driving the evolution of the price (by eliminating the terms with higher powers in Δt!)

2.2 Estimation of the Model Parameters

Following [6] and [1], we propose the following estimation method:

(i) Collect the market prices of European options for a grid of maturities τ_i and strike prices K_j.

(ii) Assuming that the model parameters were known, for given values of τ_i and K_j, an estimator of the price of the option $\widehat{V}_{ij}$ can be constructed using Monte Carlo simulations under the price model. (Is it faster to solve the dynamical system for the European price?)

(iii) Now the optimal parameters $\Theta = (\kappa, \nu, \lambda, \eta, \rho)$ may be found by solving the optimization problem:

$$\widehat{\Theta} = \text{argmin}_{\Theta} MSE(V_{ij}, \widehat{V}_{ij})$$

where V_{ij} is the market value of the option under the maturity τ_i and strike K_j.

Relate to the volatility smile

3 Pricing Options Under the LLH Model

We assume the risk free rate r to be fixed, otherwise r should be replaced by the discount factor $B(t,T) = \exp\left(-\int_t^T r_s ds\right)$. We denote by Q the martingale measure underlying the LLH model (1) - (3).

3.1 The European Price Formula

Proposition 3.1. The time t price U of a European call with spot S_t, strike K, and maturity date T is a weighted difference between the spot price S_t and the discounted strike price K

$$U(t, T, r, K, S_t) = p_1 S_t - p_2 K e^{-r(T-t)},$$

where the weights p_1 and p_2 can be obtained as probabilities of $S_T > K$ under risk neutral measures.

Proof. The time t price of a European call U with spot S_t, strike K, and maturity date T, is obtained as the risk neutral expectation conditional on the information at time t:

$$\begin{aligned} U(t, T, r, K, S_t) &= \mathbb{E}_t^Q[e^{-r(T-t)} \max(S(T) - K, 0)] \\ &= \mathbb{E}_t^Q[e^{-r(T-t)}(S(T) - K)1_{\{S_T > K\}}] \\ &= \mathbb{E}_t^Q[e^{-r(T-t)} S(T) 1_{\{S_T > K\}}] - K e^{-r(T-t)} \mathbb{E}_t^Q[1_{\{S_T > K\}}] \end{aligned}$$

To simplify the computation of the first expectation, change the numeraire to $S(t)$ and the martingale measure to Q_1 whose Radon-Nikodym derivative is given by

$$\frac{dQ_1}{dQ} = e^{-r(T-t)} \frac{S(T)}{S(t)} \tag{19}$$

Thus,

$$\begin{aligned} U(t, T, r, K, S_t, \sigma_t, \theta_t) &= \mathbb{E}_t^{Q_1}[S(t) 1_{\{S_T > K\}}] - K e^{-r(T-t)} \mathbb{E}_t^Q[1_{\{S_T > K\}}] \\ &= S(t) Q_1[S_T > K \,|S_t] - K e^{-r(T-t)} Q[S_T > K \,|S_t] \end{aligned}$$

□

Theorem 3.2. Under the dynamics (1), (2), (3), the price U of a European call U with spot S_t, strike K, and maturity date T, can be obtained as

$$U(t, T, r, K, S_t, \sigma_t, \theta_t) = S_t P_1 - K e^{-r(T-t)} P_2$$

where

$$P_j = \tfrac{1}{2} + \frac{1}{\pi} \int_0^\infty Re \left[\frac{e^{-i\phi \ln K}}{i\phi} f_j \right] d\phi \quad \text{for } j = 1, 2,$$

$f_1(\phi) = \mathbb{E}_t^{Q_1}[e^{i\phi x_T}]$, and $f_2(\phi) = \mathbb{E}_t^{Q}[e^{i\phi x_T}]$ are the characteristic functions of $x_T = \ln S_T$ conditional on the information available at time t.

Proof. Following a similar argument in the proof of the preceding proposition,

$$\begin{aligned} U(t, T, r, K, S_t, \sigma_t, \theta_t) &= \mathbb{E}_t^{Q_1}[S(t) 1_{\{S_T > K\}}] - K e^{-r(T-t)} \mathbb{E}_t^{Q}[1_{\{S_T > K\}}] \\ &= S(t) Q_1[x_T > \ln K \,|S_t] - K e^{-r(T-t)} Q[x_T > \ln K \,|S_t] \\ &= S(t) F^{Q_1}(\ln K) - K e^{-r(T-t)} F^{Q}(\ln K) \\ &= S_t P_1 - K e^{-r(T-t)} P_2, \end{aligned}$$

where $x_t = \ln S_t$, F^{Q_1} and F^Q are the distribution functions of x_T under Q_1 and Q respectively, conditional on the information available at time t. The diffusion equation of x_t is given in (9). Using the inversion theorem in [7], the probabilities P_1 and P_2 may be recovered from the respective conditional characteristic functions $f_1(\phi)$ and $f_2(\phi)$ of x_T under Q_1 and Q. By definition, $f_1(\phi) = \mathbb{E}_t^{Q_1}[e^{i\phi x_T}]$ and $f_2(\phi) = \mathbb{E}_t^{Q}[e^{i\phi x_T}]$ and the inversion theorem [7] yields

$$P_j = \tfrac{1}{2} + \frac{1}{\pi} \int_0^\infty Re \left[\frac{e^{-i\phi \ln K}}{i\phi} f_j \right] d\phi$$

□

Following the result of Theorem 3.2, **the price of a European call under the improved Stein-Stein dynamics (1), (2), (3) can be found once the characteristics f_j have been computed**. We have gleaned **two approaches** to compute the f_j's. One that is **PDE based** following **Heston** [4], and the other is **stochastic calculus** based following **Schobel and Zhu** [3]. In [1], the authors follow the former approach which using a standard martingale argument, a PDE for U is derived that ultimately yields PDEs for the $P_j's$ and consequently the $f_j's$. Following the ansatz from Lord and Kahl [5] that for affine diffusion models the $f_j's$ must be exponential in the state variables, leads to the system of ODEs described in [1] and that may be found in section 4 of this document.

In what follows we attempt a derivation of the formulas for the characteristics f_j, and consequently the European price, following the approach of Schobel and Zhu in [3]. We apply stochastic calculus machinery to derive the $f_j(\phi)$'s in terms of the model variables and parameters. We use the following lemmas in the derivation:

Lemma 3.3. If W_t^1 and W_t^2 are two correlated Brownian motions with $dW_t^1 dW_t^2 = \rho dt$, then the following representation is obtained

$$dW_t^1 = \rho dW_t^2 + \sqrt{1-\rho^2} dW_t^3$$

where $dW_t^3 dW_t^1 = \sqrt{1-\rho^2}$ and $dW_t^3 dW_t^2 = 0$.

Lemma 3.4. If H_t is non-anticipative and independent of the Brownian motion W_t, then

$$\mathbb{E}_t^Q \left[\exp\left(\alpha \int_t^T H_u dW_u \right) \right] = \exp\left(\frac{1}{2}\alpha^2 \int_t^T H_u^2 du \right) \qquad (\alpha \in \mathbb{C})$$

(add ref other than [3] and formula (4.4.31) in Shreeve's book [8])

Lemma 3.5. An Ito process following the dynamics (2) satisfies

$$\int_t^T \sigma_u dW_u^2 = \frac{1}{2\nu}\left(\sigma_T^2 - \sigma_t^2 - \nu^2(T-t) - 2\kappa\int_t^T \sigma_u\theta_u du + 2\kappa\int_t^T \sigma_u^2 du\right)$$

Proof. Applying Ito's lemma to $v_t = \sigma_t^2$ yields:

$$dv_t = 2\kappa\sigma_t\theta_t dt - 2\kappa\sigma_t^2 dt + 2\nu\sigma_t dW_t^2 + \nu^2 dt$$

Direct integration of the above equation yields:

$$\begin{aligned}
&\sigma_T^2 - \sigma_t^2 = \nu^2(T-t) + 2\kappa\int_t^T \sigma_u\theta_u du - 2\kappa\int_t^T \sigma_u^2 du + 2\nu\int_t^T \sigma_u dW_u^2 \\
\iff &\int_t^T \sigma_u dW_u^2 = \frac{1}{2\nu}\left(\sigma_T^2 - \sigma_t^2 - \nu^2(T-t) - 2\kappa\int_t^T \sigma_u\theta_u du + 2\kappa\int_t^T \sigma_u^2 du\right) \qquad \square
\end{aligned}$$

3.2 Computation of the characterisitic $f_1(\phi)$:

Step 1: Using equations (19), (9), and lemmas 3.3, 3.4 it follows that:

$$\begin{aligned}
f_1(\phi) &= \mathbb{E}_t^{Q_1}[e^{i\phi x_T}] \\
&= \mathbb{E}_t^{Q}\left[\exp\left(-r(T-t) + (1+i\phi)x_T - x_t\right)\right] \\
&= e^{i\phi(r(T-t)+x_t)}\mathbb{E}_t^{Q}\left[\exp\left\{(1+i\phi)\left(-\frac{1}{2}\int_t^T \sigma_u^2 du + \int_t^T \sigma_u dW_u^1\right)\right\}\right] \\
&= e^{i\phi(r(T-t)+x_t)}\mathbb{E}_t^{Q}\left[\exp\left\{(1+i\phi)\left(-\frac{1}{2}\int_t^T \sigma_u^2 du + \rho\int_t^T \sigma_u dW_u^2 + \sqrt{1-\rho^2}\int_t^T \sigma_u dW_u^3\right)\right\}\right] \\
&= e^{i\phi(r(T-t)+x_t)}\mathbb{E}_t^{Q}\left[\exp\left\{\frac{1+i\phi}{2}\left((1+i\phi)(1-\rho^2) - 1\right)\int_t^T \sigma_u^2 du + (1+i\phi)\rho\int_t^T \sigma_u dW_u^2\right\}\right]
\end{aligned}$$

Using lemma 3.5,

$$
\begin{aligned}
f_1(\phi) = \exp\left\{ i\phi(r(T-t)+x_t) - (1+i\phi)\frac{\rho}{2\nu}\left(\sigma_t^2 + \nu^2(T-t)\right)\right\} \times \Big(\\
\mathbb{E}_t^Q\left[\exp\left\{\frac{1+i\phi}{2}\left((1+i\phi)(1-\rho^2) - 1 + 2\frac{\kappa\rho}{\nu}\right)\int_t^T \sigma_u^2 du\right\}\right] + \\
\mathbb{E}_t^Q\left[\exp\left\{(1+i\phi)\frac{\rho}{2\nu}\sigma_T^2 - (1+i\phi)\frac{\kappa\rho}{\nu}\int_t^T \sigma_u\theta_u du\right\}\right]\Big)
\end{aligned}
$$

To simplify the notation, let $\tau = T - t$ and

$$
\begin{aligned}
s_1(\phi) &= -\frac{1+i\phi}{2}\left((1+i\phi)(1-\rho^2) - 1 + 2\frac{\kappa\rho}{\nu}\right) \\
s_2(\phi) &= (1+i\phi)\frac{\kappa\rho}{\nu} \\
s_3(\phi) &= (1+i\phi)\frac{\rho}{2\nu}
\end{aligned}
$$

Thus,

$$
\begin{aligned}
&\exp\left\{ i\phi(r\tau + x_t) - (1+i\phi)\frac{\rho}{2\nu}\left(\sigma_t^2 + \nu^2\tau\right)\right\} \\
=&\exp\left\{(i\phi r - \nu^2 s_3)\tau + i\phi x_t - s_3\sigma_t^2\right\} \\
=&\exp\left(-\frac{\rho}{2\nu}\sigma_t^2 - \frac{\rho\nu}{2}\tau\right)\exp\left\{i\phi\left[(r-\rho\nu/2)\tau + x_t - \rho/(2\nu)\sigma_t^2\right]\right\},
\end{aligned}
$$

and there remains to compute the expectation

$$
\begin{aligned}
y_\phi(t,T,\sigma,\theta) &= \mathbb{E}_t^Q\left[\exp\left\{-s_1(\phi)\int_t^T \sigma_u^2 du - s_2(\phi)\int_t^T \sigma_u\theta_u du + s_3(\phi)\sigma_T^2\right\}\right] \\
&:= \mathbb{E}_t^Q\left[\exp\left\{-\int_t^T q(\mathcal{V}_u)du\right\} g(\mathcal{V}_T)\right]
\end{aligned}
$$

where $\mathcal{V} = (\sigma, \theta)$ is the state vector, $g(\mathcal{V}_T) = e^{s_3\sigma_T^2}$, and $q(\mathcal{V}) = s_1\sigma^2 + s_2\sigma\theta = \frac{1}{2}\mathcal{V}^\top S_\phi \mathcal{V}$,

where $S_\phi = \begin{bmatrix} 2s_1(\phi) & 2s_2(\phi) \\ 0 & 0 \end{bmatrix}$.

Notice that equations (2), (3), and (5) yield $d\mathcal{V}_t = \mathbf{b}(t, \mathcal{V}_t)dt + \Sigma(t, \mathcal{V}_t)d\mathbf{W}$, which implies that $\mathcal{V}_t$ is an Ito process with drift $\mathbf{b}(t, \mathcal{V}) = \mathbf{b}(\mathcal{V})$ and diffusion matrix $\Sigma(t, \mathcal{V})$ where

$$\mathbf{b}(\mathcal{V}) = \begin{bmatrix} \kappa(\theta - \sigma) \\ \lambda \end{bmatrix} = K\mathcal{V} + L$$

$$\Sigma(t, \mathcal{V}) = \begin{bmatrix} \nu & 0 \\ 0 & \theta - \theta_0 - \lambda t + \eta \end{bmatrix} = D_1(t) + (\mathbf{e}^\top \mathcal{V})E$$

$$K = \begin{bmatrix} -\kappa & \kappa \\ 0 & 0 \end{bmatrix}, \quad L = \begin{bmatrix} 0 \\ \lambda \end{bmatrix}, \quad D_1(t) = \begin{bmatrix} \nu & 0 \\ 0 & \eta - \theta_0 - \lambda t \end{bmatrix}, \quad E = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \text{ and } \mathbf{e} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Remark 3.6. The covariance matrix $\mathcal{C}(t, \mathcal{V}) = \Sigma\Sigma^\top = \text{diag}\left(\nu^2, (\theta - \theta_0 + \eta - \lambda t)^2\right)$ and may be written as

$$\mathcal{C}(t, \mathcal{V}) = D_2(t) + D_3(t)\mathcal{V}_t + \left(\mathcal{V}^\top E\mathcal{V}\right) E$$

where $D_2(t) = \text{diag}(\nu^2, a^2(t))$, $D_3(t) = \text{diag}(0, b(t))$, $a(t) = \eta - \lambda t - \theta_0$, and $b(t) = 2(\eta - \lambda t) - 2\theta_0$.

Observation: Thus, according to the results in chapter 10 of [9], the process $\mathcal{V}$ is not affine which is a deal breaker for applying the aforementioned ansatz from Lord and Kahl [10]!

Since $\mathcal{V}_t$ is an Ito diffusion, the Feynman-Kac theorem may be applied, (see chapters 7 and 8 in [11] and the backward equation as described in the lecture notes "Some applications of the backward equation" for the extension to the inhomogeneous case. For

this purpose, the generator of $\mathcal{V}$ may be expressed as

$$\begin{aligned}\mathcal{L}u &= \mathbf{b}(\mathcal{V})\cdot\nabla u + \frac{1}{2}Tr(\mathcal{C}\nabla^2 u) \quad \text{(extend 7.3.3 in Oksendal to non-autonomous SDEs)}\\ &= \kappa(\theta-\sigma)u_\sigma + \lambda u_\theta + \frac{1}{2}\nu^2 u_{\sigma\sigma} + \frac{1}{2}(\eta-\theta_0-\lambda t+\theta)^2 u_{\theta\theta}\end{aligned}$$

The Feynman-Kac theorem yields the PDE that must be satisfied by the expectation $y_\phi(t,T,\sigma,\theta)$: (see "Some applications of the backward equation")

$$\begin{aligned}&\partial_t y_\phi + \mathcal{L}y_\phi - q(\mathcal{V})y_\phi = 0, \quad y_\phi(T,\mathcal{V}) = g(\mathcal{V})\\ \iff &\partial_t y_\phi + \mathbf{b}(\mathcal{V})\cdot\nabla_{\mathcal{V}} y_\phi + \frac{1}{2}Tr(\mathcal{C}\nabla^2_{\mathcal{V}} y_\phi) - q(\mathcal{V})y_\phi = 0, \quad y_\phi(T,\mathcal{V}) = g(\mathcal{V})\\ \iff &\frac{\partial y_\phi}{\partial t} + \frac{1}{2}\nu^2\frac{\partial^2 y_\phi}{\partial\sigma^2} + \frac{1}{2}(\eta-\theta_0-\lambda t+\theta)^2\frac{\partial^2 y_\phi}{\partial\theta^2} + \kappa(\theta-\sigma)\frac{\partial y_\phi}{\partial\sigma} + \lambda\frac{\partial y_\phi}{\partial\theta} - (s_1\sigma^2 + s_2\sigma\theta)y_\phi = 0,\\ &y_\phi(T,T,\sigma,\theta) = e^{s_3\sigma^2}\end{aligned} \tag{20}$$

Following [12], [13], and [5], for an affine diffusion the solution of the boundary value problem (20) may be conjectured as

$$y_\phi(t,\mathcal{V}) = \exp\left(\alpha_\phi(t) + \beta_\phi(t)^\top\mathcal{V} + \frac{1}{2}\mathcal{V}^\top\Gamma_\phi(t)\mathcal{V}\right) \tag{21}$$

where the coefficients $\alpha_\phi(t)$, $\beta_\phi(t)$, and $\Gamma_\phi(t)$ are to be determined (I am claiming that the diffsion $\mathcal{V}$ is not affine, however, provided the above derivation I follow the same assumption in [1] to try and recover the f_1 under the hypothesis (21))

Plugging (21) into the left hand side of following identities into (20), a system of ODEs can be derived for these coefficients. Here follows a symbolic computation of the left hand side of (20):

$$\frac{t^2\lambda^2\,\Gamma_{22}(t)}{2}+\frac{t^2\lambda^2\,\beta_2{}^2(t)}{2}-t\eta\lambda\,\Gamma_{22}(t)-t\eta\lambda\,\beta_2{}^2(t)+t\theta_0\lambda\,\Gamma_{22}(t)+t\theta_0\lambda\,\beta_2{}^2(t)+\frac{\eta^2\,\Gamma_{22}(t)}{2}+\frac{\eta^2\,\beta_2{}^2(t)}{2}-\eta\theta_0\,\Gamma_{22}(t)-\eta\theta_0\,\beta_2{}^2(t)+\frac{\theta^4\,\Gamma_{22}{}^2(t)}{2}+\theta^3\sigma\,\Gamma_{12}(t)\,\Gamma_{22}(t)+\theta^3\left(-t\lambda\,\Gamma_{22}{}^2(t)+\eta\,\Gamma_{22}{}^2(t)-\theta_0\,\Gamma_{22}{}^2(t)+\Gamma_{22}(t)\,\beta_2(t)\right)+\theta^2\sigma^2\,\Gamma_{12}{}^2(t)+$$
$$\theta^2\sigma\left(-2t\lambda\,\Gamma_{12}(t)\,\Gamma_{22}(t)+2\eta\,\Gamma_{12}(t)\,\Gamma_{22}(t)-2\theta_0\,\Gamma_{12}(t)\,\Gamma_{22}(t)+\Gamma_{12}(t)\,\beta_2(t)\right)+$$
$$\theta^2\left(\frac{t^2\lambda^2\,\Gamma_{22}{}^2(t)}{2}-t\eta\lambda\,\Gamma_{22}{}^2(t)+t\theta_0\lambda\,\Gamma_{22}{}^2(t)-2t\lambda\,\Gamma_{22}(t)\,\beta_2(t)+\frac{\eta^2\,\Gamma_{22}{}^2(t)}{2}-\eta\theta_0\,\Gamma_{22}{}^2(t)+2\eta\,\Gamma_{22}(t)\,\beta_2(t)+\frac{\theta_0^2\,\Gamma_{22}{}^2(t)}{2}-2\theta_0\,\Gamma_{22}(t)\,\beta_2(t)+\kappa\,\Gamma_{12}(t)+\frac{\Gamma_{22}(t)}{2}+\frac{\beta_2{}^2(t)}{2}+\frac{\frac{d}{dt}\,\Gamma_{22}(t)}{2}\right)+\theta\sigma^3\,\Gamma_{11}(t)\,\Gamma_{12}(t)+$$
$$\theta\sigma^2\left(-t\lambda\,\Gamma_{12}{}^2(t)+\eta\,\Gamma_{12}{}^2(t)-\theta_0\,\Gamma_{12}{}^2(t)+\Gamma_{12}(t)\,\beta_1(t)\right)+$$
$$\theta\sigma\left(-s_2+t^2\lambda^2\,\Gamma_{12}(t)\,\Gamma_{22}(t)-2t\eta\lambda\,\Gamma_{12}(t)\,\Gamma_{22}(t)+2t\theta_0\lambda\,\Gamma_{12}(t)\,\Gamma_{22}(t)-2t\lambda\,\Gamma_{12}(t)\,\beta_2(t)+\eta^2\,\Gamma_{12}(t)\,\Gamma_{22}(t)-2\eta\theta_0\,\Gamma_{12}(t)\,\Gamma_{22}(t)+2\eta\,\Gamma_{12}(t)\,\beta_2(t)+\theta_0^2\,\Gamma_{12}(t)\,\Gamma_{22}(t)-2\theta_0\,\Gamma_{12}(t)\,\beta_2(t)+\kappa\,\Gamma_{11}(t)-\kappa\,\Gamma_{12}(t)+\frac{d}{dt}\,\Gamma_{12}(t)\right)+$$
$$\theta\left(t^2\lambda^2\,\Gamma_{22}(t)\,\beta_2(t)-2t\eta\lambda\,\Gamma_{22}(t)\,\beta_2(t)+2t\theta_0\lambda\,\Gamma_{22}(t)\,\beta_2(t)-t\lambda\,\Gamma_{22}(t)-t\lambda\,\beta_2{}^2(t)+\eta^2\,\Gamma_{22}(t)\,\beta_2(t)-2\eta\theta_0\,\Gamma_{22}(t)\,\beta_2(t)+\eta\,\Gamma_{22}(t)+\eta\,\beta_2{}^2(t)+\theta_0^2\,\Gamma_{22}(t)\,\beta_2(t)-\theta_0\,\Gamma_{22}(t)-\theta_0\,\beta_2{}^2(t)+\kappa\,\beta_1(t)+\lambda\,\Gamma_{22}(t)+\frac{d}{dt}\,\beta_2(t)\right)+\frac{\theta_0^2\,\Gamma_{22}(t)}{2}+\frac{\theta_0^2\,\beta_2{}^2(t)}{2}+\lambda\,\beta_2(t)+$$
$$\frac{\sigma^4\,\Gamma_{11}{}^2(t)}{2}+\sigma^3\,\Gamma_{11}(t)\,\beta_1(t)+\sigma^2\left(-s_1+\frac{t^2\lambda^2\,\Gamma_{12}{}^2(t)}{2}-t\eta\lambda\,\Gamma_{12}{}^2(t)+t\theta_0\lambda\,\Gamma_{12}{}^2(t)+\frac{\eta^2\,\Gamma_{12}{}^2(t)}{2}-\eta\theta_0\,\Gamma_{12}{}^2(t)+\frac{\theta_0^2\,\Gamma_{12}{}^2(t)}{2}-\kappa\,\Gamma_{11}(t)+\frac{\Gamma_{11}(t)}{2}+\frac{\beta_1{}^2(t)}{2}+\frac{\frac{d}{dt}\,\Gamma_{11}(t)}{2}\right)+$$
$$\sigma\left(t^2\lambda^2\,\Gamma_{12}(t)\,\beta_2(t)-2t\eta\lambda\,\Gamma_{12}(t)\,\beta_2(t)+2t\theta_0\lambda\,\Gamma_{12}(t)\,\beta_2(t)+\eta^2\,\Gamma_{12}(t)\,\beta_2(t)-2\eta\theta_0\,\Gamma_{12}(t)\,\beta_2(t)+\theta_0^2\,\Gamma_{12}(t)\,\beta_2(t)-\kappa\,\beta_1(t)+\lambda\,\Gamma_{12}(t)+\frac{d}{dt}\,\beta_1(t)\right)+\frac{d}{dt}\alpha(t)$$

Setting the coffecients of $\sigma^i\theta^j$ equal to 0 results in a contradiction with the hypothesis (21), implying thatthere seems to be a problem with exponential of a quadratic assumption and possibly an erroneous result in [1]

4 Computation of the European Price

We describe an approach to compute the European price U using the "exact" formula presented in [1], (and attempt to inspect the bias it introduces).

The price of U depends on quantities $P_j(\tau, K)$ and $f_j(\ln S_t,\, \sigma_t,\, \theta_t, , \tau)$, $j = 1, 2$.

The f_j's depend on the quantities $D(\tau, \phi)$, $E(\tau, \phi)$, $F(\tau, \phi)$, $G(\tau, \phi)$, $H(\tau, \phi)$, and $C(\tau, \phi)$ for their evaluation. The latter parameters are defined as the solutions on the initial value problem (22).

More precisely:

$$U(\tau; S_t, K) = S_t\, P_1(\tau; K) - K e^{-r\tau}\, P_2(\tau; K)$$

where for $j = 1, 2$:

$$P_j(\tau; K) = \tfrac{1}{2} + \frac{1}{\pi}\int_0^\infty Re\left[\frac{e^{-i\phi\ln K}}{i\phi}\, f_j(\tau;\phi)\right] d\phi$$

$$f_j(\tau;\phi) = \exp\big(C + D\, v_t^2 + E\, v_t\theta_t + F\, \theta_t^2 + G\, \theta_t + H\, v_t + i\phi\, x\big)$$

$$
\begin{aligned}
\frac{\partial D}{\partial \tau}(\tau;\phi) &= \left(u_j\, i\phi - \tfrac{1}{2}\phi^2\right) + 2\nu^2 D^2(\tau;\phi) - 2A(\phi)\, D(\tau;\phi) + \tfrac{1}{2}\eta^2 E^2(\tau;\phi),\\
\frac{\partial E}{\partial \tau}(\tau;\phi) &= 2\kappa\, D(\tau;\phi) - A(\phi)\, E(\tau;\phi) + 2\nu^2 D(\tau;\phi)E(\tau;\phi) + 2\eta^2 E(\tau;\phi)F(\tau;\phi),\\
\frac{\partial F}{\partial \tau}(\tau;\phi) &= \kappa\, E(\tau;\phi) + \tfrac{1}{2}\nu^2 E^2(\tau;\phi) + 2\eta^2 F^2(\tau;\phi),\\
\frac{\partial G}{\partial \tau}(\tau;\phi) &= \kappa\, H(\tau;\phi) + 2\lambda\, F(\tau;\phi) + \nu^2 E(\tau;\phi)H(\tau;\phi) + 2\eta^2 F(\tau;\phi)G(\tau;\phi),\\
\frac{\partial H}{\partial \tau}(\tau;\phi) &= -A(\phi)\, H(\tau;\phi) + \lambda\, E(\tau;\phi) + 2\nu^2 D(\tau;\phi)H(\tau;\phi) + \eta^2 E(\tau;\phi)G(\tau;\phi),\\
\frac{\partial C}{\partial \tau}(\tau;\phi) &= r\, i\phi + \lambda\, G(\tau;\phi) + \tfrac{1}{2}\nu^2 H^2(\tau;\phi) + \nu^2 D(\tau;\phi) + \tfrac{1}{2}\eta^2 G^2(\tau;\phi) + \eta^2 F(\tau;\phi)
\end{aligned}
\tag{22}
$$

with $u_1 = \frac{1}{2}$, $u_2 = -\frac{1}{2}$, $b_1 = 1$, $b_2 = 0$, $A(\phi) = \kappa - \nu\rho\,(b_j + i\phi)$, $x = \ln S_t$ and the initial conditions $C(0;\phi) = D(0;\phi) = E(0;\phi) = F(0;\phi) = G(0;\phi) = H(0;\phi) = 0$. Here, $\tau = T - t$ is the remaining time until maturity of the option and ϕ is a dummy (frequency) parameter.

In practice, the European price $U(S_t, \sigma_t, \theta_t, t)$ may be estimated using a Monte Carlo simulation of the price process as described in section 2.1. Notice that also U may be computed numerically by integrating the system (22) as follows:

(i) Write the dynamical system in [1] involving D, E, F, G, H, and C as

$$\frac{dX}{d\tau} = f_\Theta(\tau, \phi, X)$$

where $X^\top = [D\ E\ F\ G\ H\ C]$ and $\Theta = (\kappa, \nu, \lambda, \eta, \rho)$.

(ii) For given values of Θ the dynamical system may be integrated numerically to estimate values of D, E, F, G, H, and C on a $(\tau, \phi)-$grid.

(iii) The estimated values of D, E, F, G, H, and C can be used to compute the f_j's then

integrate the P_j's, which in turn yield the value the option.

Remark 4.1.

The European price $U(S_t, \sigma_t, \theta_t, t)$ will need to be evaluated along the paths generated to evaluate the American option. The price may be estimated using a Monte Carlo simulation of the price process using the same paths to value the American option if the European option were used as a control variate (see [14] where this approach is applied to the European counterpart of the American-Bermuda-Asian option which is an option on the average stock price and can only be exercised at the final option maturity date.)

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